Vandana Sharma

Junior AI Engineer — Data Scientist

+91 7348482348 vandana.sharma727272@gmail.com GitHub — LinkedIn — Portfolio

Professional Summary

AI Engineer with hands-on experience building **LLM-powered applications, RAG pipelines, and end-to-end ML systems**. Skilled in **Python, FastAPI, and vector databases**, with practical exposure to **Generative AI, prompt engineering, and agent-based workflows**. Strong foundation in deploying scalable AI solutions on cloud platforms.

Technical Skills

Programming: Python, SQL
Machine Learning: Regression, Classification, Clustering, Feature Engineering, Model Tuning, Cross-Validation
Deep Learning & NLP: LSTM, Neural Networks, Text Processing, Sequence Models
Generative AI: LLMs, RAG, Prompt Engineering, LangChain, Agentic Workflows, Vector Databases (FAISS)
Backend & APIs: FastAPI, Flask, REST APIs
MLOps & Tools: Docker, Git, GitHub Actions, CI/CD, AWS (EC2, S3), MongoDB

Experience

Data Science Intern – Boston Consulting Group (BCG) — Forage **Feb 2025 – May 2025**

- Performed **customer segmentation** using clustering, improving targeting strategies.
- Conducted **EDA on large datasets**, identifying key behavioral patterns.
- Delivered **business insights** aligned with consulting use cases.

Machine Learning Intern – Ekak Innovation Pvt. Ltd. **Dec 2024 – Jan 2025**

- Built churn model improving accuracy by **12%**.
- Applied **feature engineering + hyperparameter tuning** for model optimization.
- Developed **model-ready pipelines** for business integration.

Data Science Intern – Earth5R **Jun 2024 – Sep 2024**

- Analyzed **150K+ records** to uncover sustainability trends.
- Built **interactive dashboards** for data storytelling.
- Generated insights supporting **impact-driven decisions**.

Data Science Intern – YBI Foundation **Jan 2024 – Feb 2024**

- Processed **200K+ unstructured records** using ETL pipelines.
- Improved **data quality and preprocessing efficiency**.

Projects

Enterprise GenAI Assistant & RAG Pipeline (Ongoing) GitHub

- Built a **GenAI-powered assistant** using LLM APIs for querying enterprise documents via semantic search.
- Designed and implemented a **Retrieval-Augmented Generation (RAG)** pipeline using **FAISS vector database** and embedding models for contextual retrieval.
- Engineered **prompt templates and agentic workflows** to handle multi-step queries and improve response consistency.
- Developed **FastAPI services** with authentication, logging, and rate limiting to expose scalable AI endpoints.

Vehicle Insurance Risk Prediction – MLOps Pipeline GitHub

- Designed **modular end-to-end ML pipeline** (data ingestion, validation, transformation, training, deployment) using configuration-driven architecture.
- Implemented **Docker-based containerization** and automated workflows via **GitHub Actions (CI/CD)** for reproducible deployments.
- Built and deployed **REST APIs using Flask/FastAPI** on AWS EC2 for real-time predictions.
- Integrated logging and exception handling mechanisms to improve pipeline reliability and debugging.

Credit Risk Prediction System (AUC: 0.93) GitHub

- Developed a complete **ML experimentation pipeline** including preprocessing, feature engineering, and imbalance handling (SMOTE).
- Compared multiple models and optimized performance using **GridSearchCV and Stratified K-Fold cross-validation**.
- Built an **ensemble stacking model** achieving AUC of 0.93 with improved generalization.
- Evaluated model performance using precision-recall trade-offs and business-relevant metrics.

Multivariate Time-Series Forecasting using LSTM GitHub

- Built a **deep learning pipeline** using LSTM networks for multivariate time-series forecasting on 1,800+ observations.
- Engineered **sliding window sequences** and optimized training using Adam optimizer and validation monitoring.
- Applied **dropout regularization** to reduce overfitting and improve model generalization.
- Performed **error propagation analysis** to evaluate prediction degradation across forecasting horizons.

Customer Churn Prediction with Explainable AI GitHub

- Built classification models with advanced preprocessing including outlier handling, transformation, and feature selection.
- Applied **SHAP explainability** to interpret model predictions and identify key churn drivers.
- Performed exploratory data analysis to uncover behavioral patterns influencing churn.
- Generated actionable insights to support **data-driven customer retention strategies**.

Academic Contributions

Research Assistance in AI/ML

- Conducted literature review and supported research documentation.
- Assisted in data analysis and interpretation for research projects.

Education

B.Tech – Computer Science & Engineering **Nov 2021 – Oct 2025**
Chhatrapati Shahu Ji Maharaj University, Kanpur

Certifications

LLMs – Google Cloud (Ongoing)
Data Science – Cisco (2024)
Machine Learning – Simplilearn (2024)